CSE 144 Final Project Report

Transfer Learning Challenge

Group Members (max 3):
Member 1 Name (CruzID: XXXXX)
Member 2 Name (CruzID: XXXXX)
Member 3 Name (CruzID: XXXXX)

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1 Introduction

1. Problem goal and setting.
2. Why transfer learning is appropriate.
3. Brief summary of your approach and main result.

2 Dataset

1. Number of classes and dataset sizes (train/val/test).
2. Directory structure and label mapping details (ensure labels 0–99 match folders).
3. Preprocessing (resize, normalization) and data augmentation used.

3 Implementation

3.1 Model

1. Pretrained backbone used (e.g., ResNet/EfficientNet/ViT) and why.
2. Architecture changes (classifier head, pooling, dropout, etc.).
3. Fine-tuning strategy (frozen layers, unfreezing schedule).

3.2 Training

1. Loss function and optimizer.
2. Learning rate, scheduler, batch size, epochs, weight decay.
3. Hardware/software environment (GPU, PyTorch version).

4 Experiments

1. Baseline setup.
2. Hyperparameter tuning plan and validation method.
3. Ablations (augmentations, model size, freezing strategy, LR, etc.).

5 Results

1. Training/validation accuracy (and loss) and any key curves/plots (optional).
2. Kaggle public leaderboard score (and submission settings).
3. Brief qualitative examples or error analysis highlights (optional).

6 Discussion

1. What worked best and why.
2. Failure cases, overfitting/underfitting observations.
3. Limitations and concrete next improvements.

7 Reproducibility

1. Random seeds and determinism settings.
2. Package versions / environment setup steps.
3. Exact commands to train and to generate `submission.csv`.

8 Team Contributions

1. Member 1: ...
2. Member 2: ...
3. Member 3: ...

9 References

1. TorchVision documentation / pretrained model sources.
2. Any papers or tutorials used.